Yu-Hsiang **Wang**

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Education

University of Illinois Urbana-Champaign — *Champaign, IL*

Ph.D. in Electrical & Computer Engineering — *Aug. 2024 - May. 2029 (expected)*

- **Advisor :** Olgica Milenkovic
- **Overall GPA :** 4.00/4.00

National Taiwan University — *Taipei, Taiwan*

B.S. in Electrical Engineering — *Sep. 2019 - Jan. 2024*

- **Overall GPA :** 4.09/4.30

Research Experience

University of Illinois Urbana-Champaign — *Champaign, IL*

Graduate Research Assistant, Advisor: Prof. Olgica Milenkovic — *Aug. 2024 - Present*

- **Time Series Generation:** Proposed WaveletDiff, a multiresolution diffusion model for time series data generation via Discrete Wavelet Transforms. [TMLR]
- **Koopman-Based Transition Path Sampling:** Formulated transition path sampling as exit-time stochastic optimal control via Koopman operators, using a single learned spectral object to sample transitions between all metastable state pairs. [NeurIPS 2026 DynaFront Workshop]
- **Robust Transition Path Sampling:** Designed stochastic control policies utilizing probabilistic bias fields to overcome initialization sensitivity in molecular transition path sampling. [NeurIPS 2026 DynaFront Workshop]
- **Molecule Generation:** Proposed DMol, a schedule-driven graph diffusion model for highly efficient and versatile molecule generation. [NeurIPS 2025]
- **Nanopore Signal Processing:** Developed a machine-learning framework utilizing wavelet feature extraction to accurately classify Abl kinase inhibitor binding types from single-molecule nanopore tweezers.

University of California, Los Angeles — *Los Angeles, CA*

Research Assistant, Advisor: Prof. Cho-Jui Hsieh — *Jun. 2023 - Sep. 2023*

- **LLM Uncertainty**: Proposed CLUE, a novel concept-level uncertainty estimation framework for Large Language Models to improve reliability and output evaluation. [NeurIPS 2024 SFLLM workshop]

Publication

[1] Boya Hou*, **Yu-Hsiang Wang***, Siddharth Ambekar, Maxim Raginsky, and Olgica Milenkovic. "Transition Path Sampling Using Koopman Operators and Exit-Time Optimal Control". *NeurIPS 2026 DynaFront Workshop, Under Review*.

[2] Ly Nguyen, **Yu-Hsiang Wang**, Joshua Foster, David DeCoeur, Lan Nguyen, Bryant Wu, Olgica Milenkovic, and Min Chen. "Subtyping Abl Kinase Inhibitor Binding Modes with Machine-Learning-Enabled Super Resolution Single-Molecule Nanopore Tweezers". *bioRxiv 2026, Under Review*.

[3] Jingqian Liu, **Yu-Hsiang Wang**, Yanru Qu, and Ge Liu. "Stochastic Control Policies for Robust Molecular Transition Path Sampling". *NeurIPS 2026 DynaFront Workshop, Under Review*.

[4] **Yu-Hsiang Wang** and Olgica Milenkovic. "WaveletDiff: Multilevel Wavelet Diffusion For Time Series Generation". *TMLR*.

[5] Peizhi Niu, **Yu-Hsiang Wang**, Vishal Rana, Chetan Rupakheti, Abhishek Pandey, and Olgica Milenkovic. "DMol: A Highly Efficient and Chemical Motif-Preserving Molecule Generation Platform". *NeurIPS 2025*.

[6] **Yu-Hsiang Wang**, Andrew Bai, Che-Ping Tsai, and Cho-Jui Hsieh. "CLUE: Concept-Level Uncertainty Estimation for Large Language Models". *NeurIPS 2024 SFLLM Workshop*.

Honors & Awards

Best Presentation Award, CSL Student Conference (CSLSC) — *Feb 2026*

Research Grant for College Students, National Science and Technology Council (Taiwan) — *July 2023 - Feb 2024*

Dean's List Award, Electrical Engineering Dept. at NTU — *Fall 2020*

Teaching

ECE313: Probability with Engineering Applications — *UIUC*

Graduate Teaching Assistant — *Fall 2025*